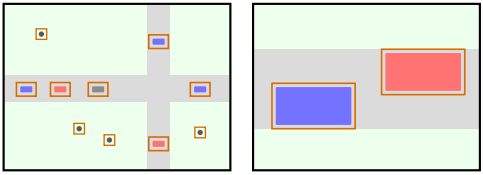


UAV camera: altitude, zoom, and position change during the flight

Scale variation

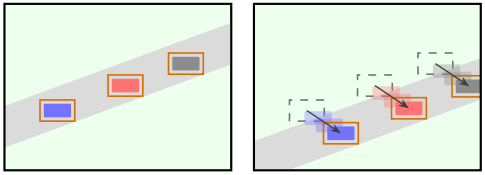


high altitude: tiny objects

low altitude: large objects

The altitude and the zoom change the size of the same objects, from a few pixels to a large part of the frame.

Camera motion



frame t

frame $t+1$
dashed: boxes of frame t

When the UAV moves, every box shifts between frames, with blur and jitter, even when the objects stand still.

Requirements for detection and tracking: reliable features and scores for small objects; data association that absorbs camera motion; real-time processing on board